MANARAT INTERNATIONAL UNIVERSITY

BUS TICKETING

PROJECT REPORT

Presented By :

Saad Al Nafis Siddiky

ID:2465CSE01176

AFM Atik Bokhtier Jarif

ID:24CSE01164

Nasrullah Khan Ferdaus

ID:2465CSE01173

Presented For :

Md Shakil Hossain

Lecturer, Dept. Of CSE

Please Turn Over/Scroll Down

BUS TICKETING PROJECT REPORT

Abstract

The Bus Ticketing System is a web-based platform designed to allow the public to purchase bus tickets from the comfort of their homes. The system provides a variety of bus types including AC, non-AC, and business class. Users can easily select their preferred bus type, view available seats, and book tickets online without needing to visit a physical counter. The platform also includes an admin panel that enables authorized administrators to manage bus listings — adding new buses or removing existing ones based on operational status or schedule changes. This system aims to simplify the ticket booking process, save time for passengers, and provide efficient fleet management for bus operators.

TABLE OF CONTENT

CONTENT	PAGE NO.
INTRODUCTION……	05
BUILDINDG OF OUR WEBSITE……	05
INTRO ABOUT WEBSITE……	05
HTML LANGUAGE……	06
PYTHON CODE……	35
OUTPUT……	45
CONCLUSION……	49

INTRODUCTION

In many countries, especially in developing regions, public transportation—particularly buses—remains the most commonly used and affordable mode of travel. However, the traditional process of purchasing bus tickets often involves visiting a physical counter, waiting in long queues, and dealing with limited availability of information regarding bus types, seat availability, and schedules. This manual system is time-consuming, inefficient, and inconvenient for passengers.

With the rapid growth of internet access and smartphone usage, there is a significant opportunity to modernize the bus ticketing process through a web-based solution. An online bus ticketing system allows passengers to browse available buses, compare different classes of service, check seat availability, and book tickets—all from the comfort of their homes or anywhere with an internet connection.

BUILDINDG OF OUR WEBSITE

The making our website needs some of the elementary things as given below:-

HTML LANGUAGE

PYTHON LANGUAGE

JAVA LANGUAGE

VISUAL STUDIO CODE TO MAKE OUR WEBSITE

GOOGLE CHROME TO RUN OUR WEBSITE

NGROC FOR HOSTING

INTRO ABOUT WEBSITE

Bus Ticket Express is a web-based bus ticketing system that allows passengers to book bus tickets online from anywhere. The website provides an easy-to-use platform where users can browse available buses, filter by bus type (All Buses, Business Class, AC Buses), and view real-time seat availability.

Users can select a bus, enter their name and phone number, choose the number of seats, and confirm their booking. The system generates a unique Booking ID for each reservation. Passengers can later search for their bookings using either the Booking ID or phone number, and cancel bookings if needed.

The website also includes an admin panel that allows administrators to add new buses or remove existing buses based on operational requirements.

HTML LANGUAGE

Hypertext markup language (HTML) is the set of markup symbols codes inserted into a file.

```
<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <meta name="viewport" content="width=device-width, initial-scale=1.0">

   <title>Bus Booking System - MySQL</title>

   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.11.0/font/bootstrap-icons.css">

   <style>

      * { margin: 0; padding: 0; box-sizing: border-box; }

      body { font-family: 'Segoe UI', sans-serif; background: #f0f2f5; }

      .nav-tabs .nav-link.active {

   background-color: #5577e6 !important;  /* Change this color */

   color: white !important;

   border-color: #5577e6 !important;

}
```

```
.nav-tabs .nav-link {
   color: #333 !important;  /* Inactive tab text color */
}
.nav-tabs .nav-link:hover {
   background-color: #b4d0fa !important;  /* Hover color */
   border-color: #b4d0fa !important;
}
      .hero { background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); color: white; padding: 60px 0; }
      .section { padding: 60px 0; }
      .bus-card { background: white; border-radius: 12px; padding: 20px; margin-bottom: 20px; box-shadow: 0 4px 6px rgba(0,0,0,0.1); transition: transform 0.2s; height: 100%; }
      .bus-card:hover { transform: translateY(-3px); box-shadow: 0 8px 15px rgba(0,0,0,0.15); }
      .bus-badge { padding: 5px 12px; border-radius: 20px; font-size: 0.75rem; font-weight: bold; }
      .badge-ac { background: #28a745; color: white; }
      .badge-business { background: #007bff; color: white; }
      .price-tag { font-size: 1.5rem; font-weight: bold; color: #007bff; }
      .btn-book { width: 100%; padding: 12px; font-weight: bold; }
      .stats-card { background: white; border-radius: 12px; padding: 20px; text-align: center; box-shadow: 0 4px 6px rgba(0,0,0,0.1); }
      .stats-number { font-size: 2.5rem; font-weight: bold; color: #007bff; }
      .booking-card { background: white; border-radius: 12px; padding: 20px; margin-bottom: 15px; box-shadow: 0 2px 4px rgba(0,0,0,0.1); }
      .toast-notification { position: fixed; top: 20px; right: 20px; z-index: 9999; min-width: 300px; }
```

```
        .db-status { background: #28a745; color: white; padding: 5px 15px; border-radius: 20px;
font-size: 0.85rem; }

        #myBookings, #adminPanel { display: none; }



    </style>

</head>

<body>

    <nav class="navbar navbar-expand-lg">

        <div class="container">

            <a class="navbar-brand" href="#" onclick="showSection('home')">

                <i class="bi bi-bus-front-fill"></i> BusTicket Express

                <span class="db-status"><i class="bi bi-database"></i> MySQL</span>

            </a>

            <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">

                <span class="navbar-toggler-icon"></span>

            </button>

            <div class="collapse navbar-collapse" id="navbarNav">

                <ul class="navbar-nav ms-auto">

                    <li class="nav-item"><a class="nav-link" href="#"
onclick="showSection('home')"><i class="bi bi-house-door"></i> Home</a></li>

                    <li class="nav-item"><a class="nav-link" href="#"
onclick="showSection('buses')"><i class="bi bi-bus-front"></i> Available Buses</a></li>

                    <li class="nav-item"><a class="nav-link" href="#"
onclick="showSection('bookings')"><i class="bi bi-ticket-perforated"></i> My
Bookings</a></li>

                    <li class="nav-item"><a class="nav-link" href="#"
onclick="showSection('admin')"><i class="bi bi-shield-lock"></i> Admin</a></li>

                </ul>
```

```
        </div>
      </div>
    </nav>

    <div id="toastContainer" class="toast-notification"></div>

    <!-- Home Section -->
    <div id="homeSection">
      <div class="hero">
        <div class="container">
          <div class="row align-items-center">
            <div class="col-lg-6">
              <h1 class="display-4 fw-bold mb-4">Book Your Bus Ticket <span style="color:
#ffd700;">Easily</span></h1>
              <p class="lead mb-4">Travel comfortably with our premium bus services.</br>
              <b>Your Journy Our Passion.</b></p>
              <button class="btn btn-light btn-lg" onclick="showSection('buses')">
                View Available Buses <i class="bi bi-arrow-right"></i>
              </button>
            </div>
            <div class="col-lg-6 text-center">
              <i class="bi bi-bus-front" style="font-size: 200px;"></i>
            </div>
          </div>
        </div>
      </div>
```

```
<div class="container section">
  <div class="row">
    <div class="col-md-4 mb-3">
      <div class="stats-card">
        <i class="bi bi-bus-front" style="font-size: 3rem; color: #007bff;"></i>
        <div class="stats-number" id="totalBuses">0</div>
        <div>Total Buses</div>
      </div>
    </div>
    <div class="col-md-4 mb-3">
      <div class="stats-card">
        <i class="bi bi-database" style="font-size: 3rem; color: #28a745;"></i>
        <div class="stats-number" id="dbStatus">✓ </div>
        <div>Database Status</div>
      </div>
    </div>
    <div class="col-md-4 mb-3">
      <div class="stats-card">
        <i class="bi bi-people" style="font-size: 3rem; color: #ffc107;"></i>
        <div class="stats-number" id="totalSeats">0</div>
        <div>Available Seats</div>
      </div>
    </div>
  </div>
</div>
```

```
</div>


<!-- Buses Section -->
<div id="busesSection" style="display: none;">
  <div class="container section">
    <h2 class="mb-4"><i class="bi bi-bus-front"></i> Available Buses</h2>
    <ul class="nav nav-tabs" id="busTabs">
      <li class="nav-item"><a class="nav-link active" data-bs-toggle="tab" href="#allBuses">All Buses</a></li>
      <li class="nav-item"><a class="nav-link" data-bs-toggle="tab" href="#businessBuses">Business Class</a></li>
      <li class="nav-item"><a class="nav-link" data-bs-toggle="tab" href="#acBuses">AC Buses</a></li>
    </ul>
    <div class="tab-content">
      <div class="tab-pane fade show active" id="allBuses">
        <div class="row mt-4" id="allBusesContainer"></div>
      </div>
      <div class="tab-pane fade" id="businessBuses">
        <div class="row mt-4" id="businessBusesContainer"></div>
      </div>
      <div class="tab-pane fade" id="acBuses">
        <div class="row mt-4" id="acBusesContainer"></div>
      </div>
    </div>
  </div>
</div>
```

```
<!-- Bookings Section -->
<div id="bookingsSection" style="display: none;">
  <div class="container section">
    <h2 class="mb-4"><i class="bi bi-ticket-perforated"></i> My Bookings</h2>
    <div class="row mb-4">
      <div class="col-md-6">
        <div class="input-group">
          <input type="text" class="form-control" id="searchBookingInput"
placeholder="Enter Booking ID or Phone Number">
          <button class="btn btn-primary" onclick="searchBookings()"><i class="bi bi-
search"></i> Search</button>
        </div>
      </div>
    </div>
    <div id="bookingsList"><p class="text-muted text-center">Enter your booking details
to view</p></div>
  </div>
</div>


<!-- Admin Section -->
<div id="adminSection" style="display: none;">
  <div class="container section">
    <h2 class="mb-4"><i class="bi bi-shield-lock"></i> Admin Panel</h2>
    <div id="adminLoginForm">
      <div class="row">
        <div class="col-md-4 mx-auto">
```

```
                <div class="card">
                    <div class="card-body">
                        <h5 class="card-title">Admin Login</h5>
                        <div class="mb-3">
                            <label class="form-label">Password</label>
                            <input type="password" class="form-control"
id="adminPasswordInput">
                        </div>
                        <button class="btn btn-primary w-100"
onclick="adminLogin()">Login</button>
                        <button class="btn btn-link w-100 mt-2" onclick="setupAdmin()">Setup
New Password</button>
                    </div>
                </div>
            </div>
        </div>
        <div id="adminDashboard" style="display: none;">
            <div class="row mb-4">
                <div class="col-md-4 mb-3">
                    <div class="stats-card">
                        <div class="stats-number" id="adminTotalBuses">0</div>
                        <div>Total Buses</div>
                    </div>
                </div>
                <div class="col-md-4 mb-3">
                    <div class="stats-card">
```

```
                <div class="stats-number" id="adminTotalBookings">0</div>
                <div>Total Bookings</div>
            </div>
        </div>
        <div class="col-md-4 mb-3">
            <div class="stats-card">
                <div class="stats-number" id="adminTotalRevenue">0</div>
                <div>Total Revenue</div>
            </div>
        </div>
    </div>
    <!-- Manage Existing Buses Card -->
<div class="card mb-4">
  <div class="card-header bg-info text-white">
    <h5><i class="bi bi-gear"></i> Manage Existing Buses</h5>
  </div>
  <div class="card-body">
    <div class="table-responsive">
      <table class="table table-hover">
        <thead>
          <tr>
            <th>Bus #</th>
            <th>Bus Name</th>
            <th>Route</th>
            <th>Type</th>
            <th>Seats</th>
```

```
                <th>Status</th>
                <th>Actions</th>
              </tr>
            </thead>
            <tbody id="manageBusesTable">
              <tr>
                <td colspan="7" class="text-center">Loading buses...</td>
              </tr>
            </tbody>
          </table>
        </div>
      </div>
    </div>
<div class="d-flex justify-content-between align-items-center mb-3">
    <h6 class="mb-0"><i class="bi bi-list"></i> Bus List</h6>
    <button class="btn btn-sm btn-outline-primary" onclick="loadManageBusesTable()">
        <i class="bi bi-arrow-clockwise"></i> Refresh
    </button>
</div>


            <div class="card mb-4">
                <div class="card-header"><h5><i class="bi bi-plus-circle"></i> Add New
Bus</h5></div>
                <div class="card-body">
                    <div class="row">
                        <div class="col-md-2 mb-2"><input type="number" class="form-control"
id="newBusNo" placeholder="Bus Number"></div>
```

```
                    <div class="col-md-2 mb-2"><input type="text" class="form-control"
id="newBusName" placeholder="Bus Name"></div>

                    <div class="col-md-2 mb-2"><input type="text" class="form-control"
id="newBusRoute" placeholder="Route"></div>

                    <div class="col-md-2 mb-2">

                      <select class="form-control" id="newBusType">

                        <option value="BUSINESS">Business</option>

                        <option value="AC">AC</option>

                      </select>

                    </div>

                    <div class="col-md-2 mb-2"><input type="number" class="form-control"
id="newBusSeats" placeholder="Total Seats"></div>

                    <div class="col-md-2 mb-2"><input type="number" class="form-control"
id="newBusRent" placeholder="Rent"></div>

                  </div>

                  <button class="btn btn-primary" onclick="addNewBus()">Add Bus</button>

                </div>

              </div>

              <button class="btn btn-danger" onclick="adminLogout()"><i class="bi bi-box-arrow-
right"></i> Logout</button>

          </div>

        </div>

    </div>


    <!-- Booking Modal -->

    <div class="modal fade" id="bookingModal" tabindex="-1">

        <div class="modal-dialog">

            <div class="modal-content">
```

```
<div class="modal-header bg-primary text-white">
  <h5 class="modal-title"><i class="bi bi-ticket-perforated"></i> Book Your Ticket</h5>
  <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal"></button>
</div>
<div class="modal-body">
  <input type="hidden" id="selectedBusNo">
  <input type="hidden" id="selectedBusRent">
  <div class="alert alert-info" id="busDetailsAlert"></div>
  <div class="mb-3">
    <label class="form-label">Full Name *</label>
    <input type="text" class="form-control" id="passengerName">
  </div>
  <div class="mb-3">
    <label class="form-label">Phone Number *</label>
    <input type="tel" class="form-control" id="passengerPhone">
  </div>
  <div class="mb-3">
    <label class="form-label">Number of Seats(Maximum 4 Seats) *</label>
    <input type="number" class="form-control" id="seatsBooked" min="1" max="4">
  </div>
  <div class="mb-3">
    <label class="form-label">Total Amount</label>
    <div id="totalAmount" class="h4 text-primary">৳</div>
  </div>
```

```
            </div>
            <div class="modal-footer">
                <button type="button" class="btn btn-secondary" data-bs-
dismiss="modal">Cancel</button>
                <button type="button" class="btn btn-primary"
onclick="confirmBooking()">Confirm Booking</button>
            </div>
        </div>
    </div>
  </div>


  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></scri
pt>
  <script>
    let adminLoggedIn = false;


    function showSection(section) {
        document.getElementById('homeSection').style.display = section === 'home' ? 'block'
: 'none';
        document.getElementById('busesSection').style.display = section === 'buses' ? 'block'
: 'none';
        document.getElementById('bookingsSection').style.display = section === 'bookings' ?
'block' : 'none';
        document.getElementById('adminSection').style.display = section === 'admin' ?
'block' : 'none';


        if (section === 'buses') loadBuses();
        if (section === 'home') loadStats();
```

```
}

async function loadStats() {
    try {
        const response = await fetch('/api/buses');
        const buses = await response.json();
        const totalSeats = buses.reduce((sum, b) => sum + b.available_seats, 0);
        document.getElementById('totalBuses').textContent = buses.length;
        document.getElementById('totalSeats').textContent = totalSeats;

        // Check DB connection
        const dbRes = await fetch('/api/health');
        const dbData = await dbRes.json();
        document.getElementById('dbStatus').textContent = dbData.database ===
'connected' ? '✓' : '✗';
    } catch (error) {
        console.error('Error:', error);
    }
}

async function loadBuses() {
    try {
        const response = await fetch('/api/buses');
        const buses = await response.json();
        displayBuses(buses.filter(b => b.available_seats >= 0), 'allBusesContainer');
        displayBuses(buses.filter(b => b.bus_type === 'BUSINESS' && b.available_seats >=
0), 'businessBusesContainer');
```

```
            displayBuses(buses.filter(b => b.bus_type === 'AC' && b.available_seats >= 0),
'acBusesContainer');
        } catch (error) {
            showToast('Error loading buses', 'error');
        }
    }

    function displayBuses(buses, containerId) {
        const container = document.getElementById(containerId);
        if (buses.length === 0) {
            container.innerHTML = '<p class="text-muted">No buses available</p>';
            return;
        }
        container.innerHTML = buses.map(bus => `
            <div class="col-md-6 col-lg-4">
                <div class="bus-card">
                    <div class="d-flex justify-content-between">
                        <h5>${bus.bus_name}</h5>
                        <span class="bus-badge ${bus.bus_type === 'AC' ? 'badge-ac' : 'badge-
business'}">${bus.bus_type}</span>
                    </div>
                    <p><i class="bi bi-geo-alt"></i> ${bus.route}</p>
                    <p><i class="bi bi-hash"></i> Bus No: ${bus.bus_no}</p>
                    <p><i class="bi bi-people"></i> Available:
${bus.available_seats}/${bus.total_seats}</p>
                    <div class="price-tag mb-3">BDT ${bus.rent}</div>
```

```
                <button class="btn btn-primary btn-book" onclick='openBookingModal(${JSON.stringify(bus)})'>Book Now</button>
            </div>
        </div>
    `).join('');
}

function openBookingModal(bus) {
    document.getElementById('selectedBusNo').value = bus.bus_no;
    document.getElementById('selectedBusRent').value = bus.rent;
    document.getElementById('busDetailsAlert').innerHTML = `<strong>${bus.bus_name}</strong><br>Route: ${bus.route}<br>Available: ${bus.available_seats} seats<br>Rent: BDT ${bus.rent}`;
    document.getElementById('seatsBooked').max = Math.min(4, bus.available_seats);
    new bootstrap.Modal(document.getElementById('bookingModal')).show();
}

document.addEventListener('DOMContentLoaded', function() {
    document.getElementById('seatsBooked').addEventListener('input', function() {
        const seats = parseInt(this.value) || 0;
        const rent = parseInt(document.getElementById('selectedBusRent').value) || 0;
        document.getElementById('totalAmount').textContent = 'BDT ' + (seats * rent);
    });
});

async function confirmBooking() {
    const data = {
```

```
        bus_no: parseInt(document.getElementById('selectedBusNo').value),
        passenger_name: document.getElementById('passengerName').value,
        passenger_phone: document.getElementById('passengerPhone').value,
        seats_booked: parseInt(document.getElementById('seatsBooked').value)
    };

    if (!data.passenger_name || !data.passenger_phone || data.seats_booked < 1) {
        showToast('Please fill all fields', 'warning');
        return;
    }

    try {
        const response = await fetch('/api/bookings', {
            method: 'POST',
            headers: {'Content-Type': 'application/json'},
            body: JSON.stringify(data)
        });
        const result = await response.json();
        if (response.ok) {
            showToast(`Booking Confirmed! ID: ${result.booking_id}`, 'success');

bootstrap.Modal.getInstance(document.getElementById('bookingModal')).hide();
            loadBuses();
        } else {
            showToast(result.error, 'error');
        }
```

```
    } catch (error) {
        showToast('Network error', 'error');
    }
}

async function searchBookings() {
    const query = document.getElementById('searchBookingInput').value;
    if (!query) return;

    try {
        const response = await fetch(`/api/bookings/search?q=${query}`);
        const bookings = await response.json();
        const container = document.getElementById('bookingsList');

        if (bookings.length === 0) {
            container.innerHTML = '<p class="text-muted text-center">No bookings
found</p>';
            return;
        }

        container.innerHTML = bookings.map(b => `
            <div class="booking-card">
                <div class="d-flex justify-content-between">
                    <h6>Booking ID: ${b.booking_id}</h6>
                    <span class="badge bg-${b.status === 'CONFIRMED' ? 'success' :
'danger'}">${b.status}</span>
                </div>
```

```
                <p><strong>Bus:</strong> ${b.bus_name} (#${b.bus_no})</p>

                <p><strong>Passenger:</strong> ${b.passenger_name} |
${b.passenger_phone}</p>

                <p><strong>Seats:</strong> ${b.seats_booked} | <strong>Amount:</strong>
BDT ${b.total_amount}</p>

                ${b.status === 'CONFIRMED' ? `<button class="btn btn-danger btn-sm"
onclick="cancelBooking(${b.booking_id})">Cancel</button>` : ''}

            </div>

        `).join('');

    } catch (error) {

        showToast('Error searching', 'error');

    }

}


async function cancelBooking(id) {

    if (!confirm('Cancel this booking?')) return;

    try {

        await fetch(`/api/bookings/${id}/cancel`, {method: 'POST'});

        showToast('Booking cancelled', 'success');

        searchBookings();

        loadBuses();

    } catch (error) {

        showToast('Error cancelling', 'error');

    }

}


async function setupAdmin() {
```

```
    const password = prompt('Enter new admin password (min 6 chars):');
    if (!password || password.length < 6) return;
    try {
        await fetch('/api/admin/setup', {
            method: 'POST',
            headers: {'Content-Type': 'application/json'},
            body: JSON.stringify({password})
        });
        showToast('Admin setup complete', 'success');
    } catch (error) {
        showToast('Setup failed', 'error');
    }
}

async function adminLogin() {
    const password = document.getElementById('adminPasswordInput').value;
    try {
        const res = await fetch('/api/admin/login', {
            method: 'POST',
            headers: {'Content-Type': 'application/json'},
            body: JSON.stringify({password})
        });
        if (res.ok) {
            adminLoggedIn = true;
            document.getElementById('adminLoginForm').style.display = 'none';
            document.getElementById('adminDashboard').style.display = 'block';
```

```
            loadAdminDashboard();
            loadManageBusesTable();
        } else {
            showToast('Invalid password', 'error');
        }
    } catch (error) {
        showToast('Login failed', 'error');
    }
}

// Load buses into management table
async function loadManageBusesTable() {
    try {
        const response = await fetch('/api/buses');
        const buses = await response.json();

        const tbody = document.getElementById('manageBusesTable');

        if (buses.length === 0) {
            tbody.innerHTML = '<tr><td colspan="7" class="text-center text-muted">No buses
found</td></tr>';
            return;
        }

        tbody.innerHTML = buses.map(bus => {
            const seatStatus = bus.available_seats === bus.total_seats ? 'full' :
```

```
                (bus.available_seats === 0 ? 'empty' : 'partial');

        const statusBadge = {
            'full': '<span class="badge bg-success">Empty</span>',
            'empty': '<span class="badge bg-danger">Full</span>',
            'partial': '<span class="badge bg-warning">Partial</span>'
        }[seatStatus];

        return `
            <tr>
                <td><strong>#${bus.bus_no}</strong></td>
                <td>${bus.bus_name}</td>
                <td>${bus.route}</td>
                <td>
                    <span class="badge ${bus.bus_type === 'AC' ? 'bg-success' : 'bg-primary'}">
                        ${bus.bus_type}
                    </span>
                </td>
                <td>
                    <strong>${bus.available_seats}</strong> / ${bus.total_seats}
                </td>
                <td>${statusBadge}</td>
                <td>
                    <button class="btn btn-sm btn-warning"
                            onclick="resetSingleBus(${bus.bus_no}, '${bus.bus_name}',
${bus.total_seats})"
```

```
                    ${bus.available_seats === bus.total_seats ? 'disabled' : ''}>
                  <i class="bi bi-arrow-repeat"></i> Reset
                </button>
                <button class="btn btn-sm btn-danger ms-1"
                    onclick="deleteBus(${bus.bus_no}, '${bus.bus_name}')">
                  <i class="bi bi-trash"></i>
                </button>
              </td>
            </tr>
          `;
      }).join('');

    } catch (error) {
      console.error('Error loading buses:', error);
      showToast('Error loading buses', 'error');
    }
}

// Reset a single bus
async function resetSingleBus(busNo, busName, totalSeats) {
    if (!confirm(`Reset Bus #${busNo} (${busName}) to ${totalSeats} available seats?`)) {
      return;
    }

    try {
      const response = await fetch(`/api/admin/reset-bus/${busNo}`, {
```

```
        method: 'POST',
        headers: {'Content-Type': 'application/json'}
    });

    const result = await response.json();

    if (response.ok) {
        showToast(result.message, 'success');
        loadManageBusesTable();  // Refresh the table
        loadAdminDashboard();     // Refresh stats
        loadBuses();              // Refresh bus listings
    } else {
        showToast(result.error || 'Failed to reset bus', 'error');
    }
  } catch (error) {
    showToast('Network error', 'error');
  }
}

// Delete a bus (only if no active bookings)
async function deleteBus(busNo, busName) {
  if (!confirm(`Are you sure you want to delete Bus #${busNo} (${busName})?\nThis action cannot be undone!`)) {
    return;
  }
```

```
    try {
        const response = await fetch(`/api/buses/${busNo}`, {
            method: 'DELETE'
        });

        const result = await response.json();

        if (response.ok) {
            showToast(`✅ Bus #${busNo} deleted successfully!`, 'success');
            loadManageBusesTable();
            loadAdminDashboard();
            loadBuses();
        } else {
            showToast(result.error || 'Cannot delete bus with active bookings', 'error');
        }
    } catch (error) {
        showToast('Network error', 'error');
    }
}

        async function loadAdminDashboard() {
    try {
        const [busesRes, bookingsRes] = await Promise.all([
            fetch('/api/buses'),
            fetch('/api/bookings')
        ]);
```

```
    const buses = await busesRes.json();
    const bookings = await bookingsRes.json();

    const confirmed = bookings.filter(b => b.status === 'CONFIRMED');

    // FIX: Convert to numbers before adding
    const revenue = confirmed.reduce((sum, b) => {
        return sum + parseFloat(b.total_amount || 0);
    }, 0);

    // Format revenue with 2 decimal places
    const formattedRevenue = revenue.toFixed(2);

    document.getElementById('adminTotalBuses').textContent = buses.length;
    document.getElementById('adminTotalBookings').textContent = bookings.length;
    document.getElementById('adminTotalRevenue').textContent = 'BDT ' +
formattedRevenue;

    // Optional: Display active bookings count
    const activeBookingsEl = document.getElementById('adminActiveBookings');
    if (activeBookingsEl) {
        activeBookingsEl.textContent = confirmed.length;
    }

  async function loadAdminDashboard() {
  try {
```

```
        const [busesRes, bookingsRes] = await Promise.all([
            fetch('/api/buses'),
            fetch('/api/bookings')
        ]);
        const buses = await busesRes.json();
        const bookings = await bookingsRes.json();

        const confirmed = bookings.filter(b => b.status === 'CONFIRMED');
        const revenue = confirmed.reduce((sum, b) => sum + parseFloat(b.total_amount || 0),
0);
        const formattedRevenue = revenue.toFixed(2);

        document.getElementById('adminTotalBuses').textContent = buses.length;
        document.getElementById('adminTotalBookings').textContent = bookings.length;
        document.getElementById('adminTotalRevenue').textContent = 'BDT ' +
formattedRevenue;

        const activeBookingsEl = document.getElementById('adminActiveBookings');
        if (activeBookingsEl) {
            activeBookingsEl.textContent = confirmed.length;
        }

        // Load the manage buses table
        loadManageBusesTable();

    } catch (error) {
        console.error('Error loading dashboard:', error);
```

```
    }
}

    } catch (error) {
        console.error('Error loading dashboard:', error);
    }
}

        async function addNewBus() {
            const data = {
                bus_no: parseInt(document.getElementById('newBusNo').value),
                bus_name: document.getElementById('newBusName').value,
                route: document.getElementById('newBusRoute').value,
                bus_type: document.getElementById('newBusType').value,
                total_seats: parseInt(document.getElementById('newBusSeats').value),
                rent: parseInt(document.getElementById('newBusRent').value)
            };

            try {
                const res = await fetch('/api/buses', {
                    method: 'POST',
                    headers: {'Content-Type': 'application/json'},
                    body: JSON.stringify(data)
                });
                if (res.ok) {
                    showToast('Bus added', 'success');
```

```
            loadAdminDashboard();
            loadBuses();
        }
    } catch (error) {
        showToast('Error adding bus', 'error');
    }
}

function adminLogout() {
    adminLoggedIn = false;
    document.getElementById('adminLoginForm').style.display = 'block';
    document.getElementById('adminDashboard').style.display = 'none';
}

function showToast(message, type) {
    const colors = {success: 'bg-success', error: 'bg-danger', warning: 'bg-warning'};
    const toast = document.createElement('div');
    toast.className = `toast show ${colors[type]} text-white`;
    toast.style.marginBottom = '10px';
    toast.innerHTML = `<div class="toast-body">${message}</div>`;
    document.getElementById('toastContainer').appendChild(toast);
    setTimeout(() => toast.remove(), 3000);
}

loadStats();
</script>
```

```
</body>
</html>
```

PYTHON CODE

```
from flask import Flask, jsonify, request, send_from_directory
from flask_cors import CORS
import pymysql
import os
from db_config import get_connection, init_database

app = Flask(__name__, static_folder='../frontend', static_url_path='')
CORS(app)

# Initialize database on startup
init_database()

# Get the absolute path to frontend folder
BASE_DIR = os.path.dirname(os.path.abspath(__file__))
FRONTEND_DIR = os.path.join(BASE_DIR, '..', 'frontend')

# ==================== ROUTES ====================
@app.route('/')
def home():
    """Serve the index.html file"""
    return send_from_directory(FRONTEND_DIR, 'index.html')
```

```
@app.route('/<path:filename>')
def serve_static(filename):
    """Serve static files from frontend folder"""
    return send_from_directory(FRONTEND_DIR, filename)


@app.route('/api/health')
def health():
    try:
        conn = get_connection()
        conn.close()
        return jsonify({"status": "healthy", "database": "connected"})
    except:
        return jsonify({"status": "healthy", "database": "disconnected"})


@app.route('/api/buses', methods=['GET'])
def get_buses():
    conn = get_connection()
    cursor = conn.cursor()
    cursor.execute("SELECT * FROM buses ORDER BY bus_no")
    buses = cursor.fetchall()
    cursor.close()
    conn.close()
    return jsonify(buses)


@app.route('/api/buses', methods=['POST'])
def add_bus():
```

```
    data = request.json
    conn = get_connection()
    cursor = conn.cursor()
    cursor.execute("""
        INSERT INTO buses (bus_no, bus_name, route, bus_type, total_seats, available_seats,
rent)
        VALUES (%s, %s, %s, %s, %s, %s, %s)
    """, (data['bus_no'], data['bus_name'], data['route'], data['bus_type'],
          data['total_seats'], data['total_seats'], data['rent']))
    conn.commit()
    cursor.close()
    conn.close()
    return jsonify({"message": "Bus added"}), 201

@app.route('/api/buses/<int:bus_no>', methods=['DELETE'])
def delete_bus(bus_no):
    """Delete a bus (only if no active bookings)"""
    conn = get_connection()
    cursor = conn.cursor()

    try:
        # Check for active bookings
        cursor.execute("""
            SELECT COUNT(*) as count
            FROM bookings
            WHERE bus_no = %s AND status = 'CONFIRMED'
```

```
        """, (bus_no,))

        result = cursor.fetchone()
        if result['count'] > 0:
            return jsonify({"error": "Cannot delete bus with active bookings"}), 400

        # Delete the bus
        cursor.execute("DELETE FROM buses WHERE bus_no = %s", (bus_no,))

        if cursor.rowcount == 0:
            return jsonify({"error": "Bus not found"}), 404

        conn.commit()
        return jsonify({"message": f"Bus #{bus_no} deleted successfully"}), 200

    except Exception as e:
        conn.rollback()
        return jsonify({"error": str(e)}), 500
    finally:
        cursor.close()
        conn.close()

@app.route('/api/bookings', methods=['GET'])
def get_bookings():
    conn = get_connection()
    cursor = conn.cursor()
```

```
    cursor.execute("""
        SELECT b.*, bs.bus_name, bs.route
        FROM bookings b
        JOIN buses bs ON b.bus_no = bs.bus_no
        ORDER BY b.booking_date DESC
    """)
    bookings = cursor.fetchall()
    cursor.close()
    conn.close()
    return jsonify(bookings)

@app.route('/api/bookings', methods=['POST'])
def create_booking():
    data = request.json
    conn = get_connection()
    cursor = conn.cursor()

    # Check maximum 4 tickets per booking
    if data['seats_booked'] > 4:
        return jsonify({"error": "Maximum 4 tickets per booking allowed!"}), 400
    if data['seats_booked'] < 1:
        return jsonify({"error": "Must book at least 1 seat!"}), 400

    cursor.execute("SELECT * FROM buses WHERE bus_no = %s", (data['bus_no'],))
    bus = cursor.fetchone()
```

```
    if not bus:
        return jsonify({"error": "Bus not found"}), 404

    if bus['available_seats'] < data['seats_booked']:
        return jsonify({"error": f"Only {bus['available_seats']} seats available!"}), 400

    total = data['seats_booked'] * bus['rent']

    cursor.execute("""
        INSERT INTO bookings (bus_no, passenger_name, passenger_phone, seats_booked,
total_amount)
        VALUES (%s, %s, %s, %s, %s)
    """, (data['bus_no'], data['passenger_name'], data['passenger_phone'],
          data['seats_booked'], total))

    booking_id = cursor.lastrowid

    cursor.execute("UPDATE buses SET available_seats = available_seats - %s WHERE bus_no
= %s",
                   (data['seats_booked'], data['bus_no']))

    conn.commit()
    cursor.close()
    conn.close()

    return jsonify({
        "booking_id": booking_id,
```

```
        "message": f"Booking confirmed! {data['seats_booked']} seat(s) booked successfully."
    }), 201

@app.route('/api/bookings/search')
def search_bookings():
    query = request.args.get('q', '')
    conn = get_connection()
    cursor = conn.cursor()
    cursor.execute("""
        SELECT b.*, bs.bus_name, bs.route
        FROM bookings b
        JOIN buses bs ON b.bus_no = bs.bus_no
        WHERE b.booking_id LIKE %s OR b.passenger_phone LIKE %s
        ORDER BY b.booking_date DESC
    """, (f'%{query}%', f'%{query}%'))
    bookings = cursor.fetchall()
    cursor.close()
    conn.close()
    return jsonify(bookings)

@app.route('/api/bookings/<int:booking_id>/cancel', methods=['POST'])
def cancel_booking(booking_id):
    conn = get_connection()
    cursor = conn.cursor()
```

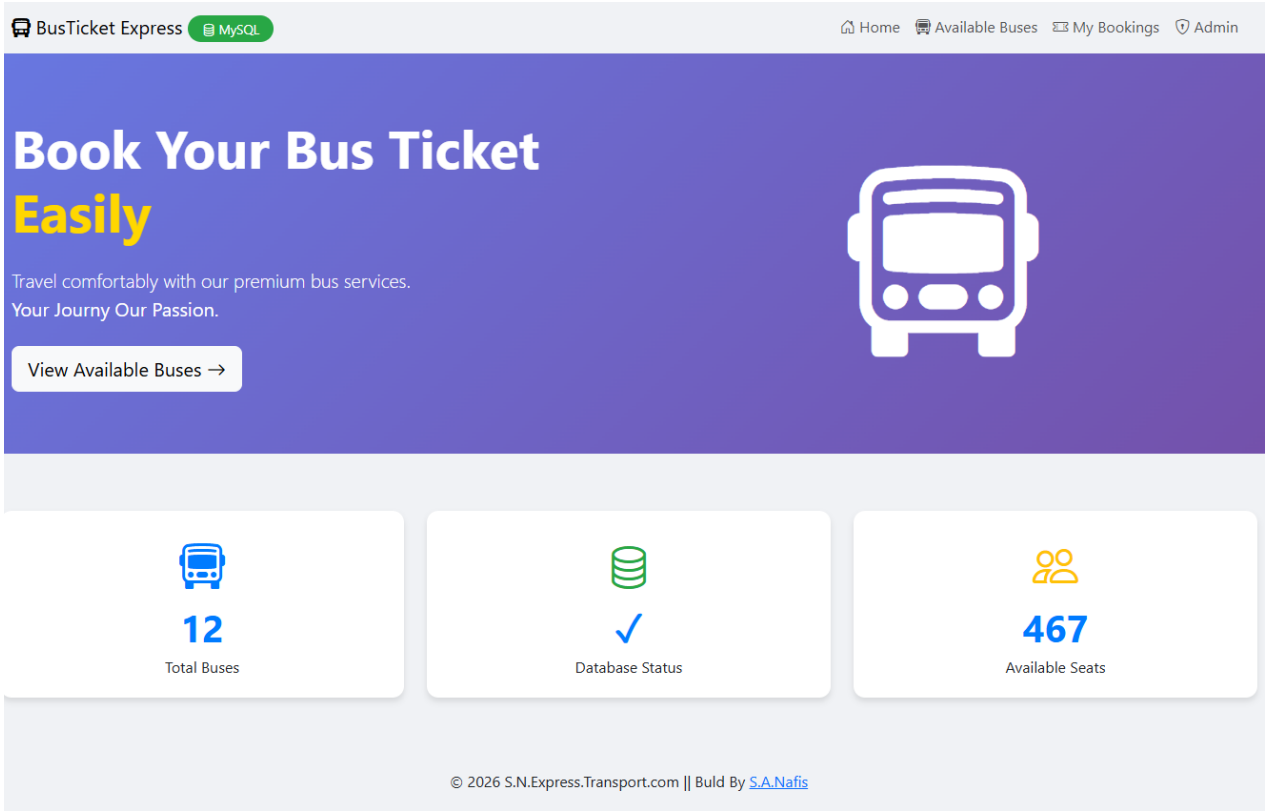
```
    cursor.execute("SELECT * FROM bookings WHERE booking_id = %s AND status =
'CONFIRMED'", (booking_id,))
    booking = cursor.fetchone()

    if booking:
        cursor.execute("UPDATE bookings SET status = 'CANCELLED' WHERE booking_id = %s",
(booking_id,))
        cursor.execute("UPDATE buses SET available_seats = available_seats + %s WHERE
bus_no = %s",
                       (booking['seats_booked'], booking['bus_no']))
        conn.commit()

    cursor.close()
    conn.close()
    return jsonify({"message": "Cancelled"})

@app.route('/api/admin/setup', methods=['POST'])
def admin_setup():
    data = request.json
    conn = get_connection()
    cursor = conn.cursor()
    cursor.execute("SELECT COUNT(*) as count FROM admins")
    if cursor.fetchone()['count'] > 0:
        return jsonify({"error": "Admin already exists"}), 400

    cursor.execute("INSERT INTO admins (password) VALUES (%s)", (data['password'],))
    conn.commit()
```

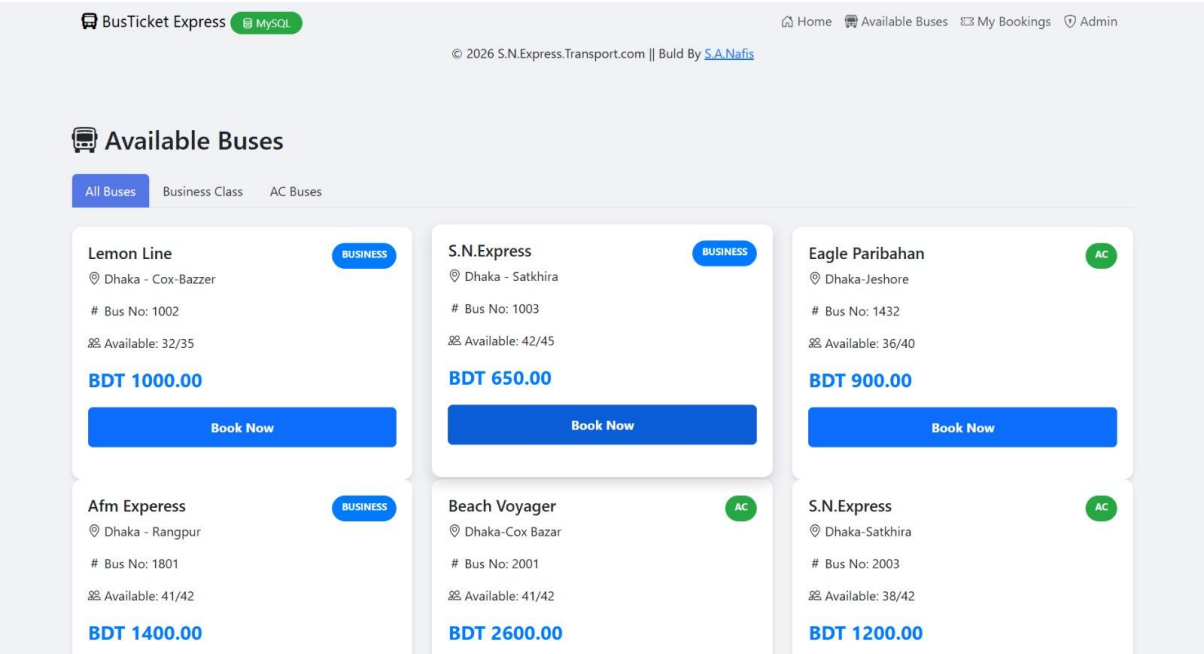
```
    cursor.close()
    conn.close()
    return jsonify({"message": "Setup complete"})

@app.route('/api/admin/login', methods=['POST'])
def admin_login():
    data = request.json
    conn = get_connection()
    cursor = conn.cursor()
    cursor.execute("SELECT * FROM admins WHERE password = %s", (data['password'],))
    admin = cursor.fetchone()
    cursor.close()
    conn.close()

    if admin:
        return jsonify({"message": "Login successful"})
    return jsonify({"error": "Invalid password"}), 401

@app.route('/api/admin/reset-bus/<int:bus_no>', methods=['POST'])
def reset_single_bus(bus_no):
    """Reset available seats for a specific bus"""
    conn = get_connection()
    cursor = conn.cursor()

    try:
```

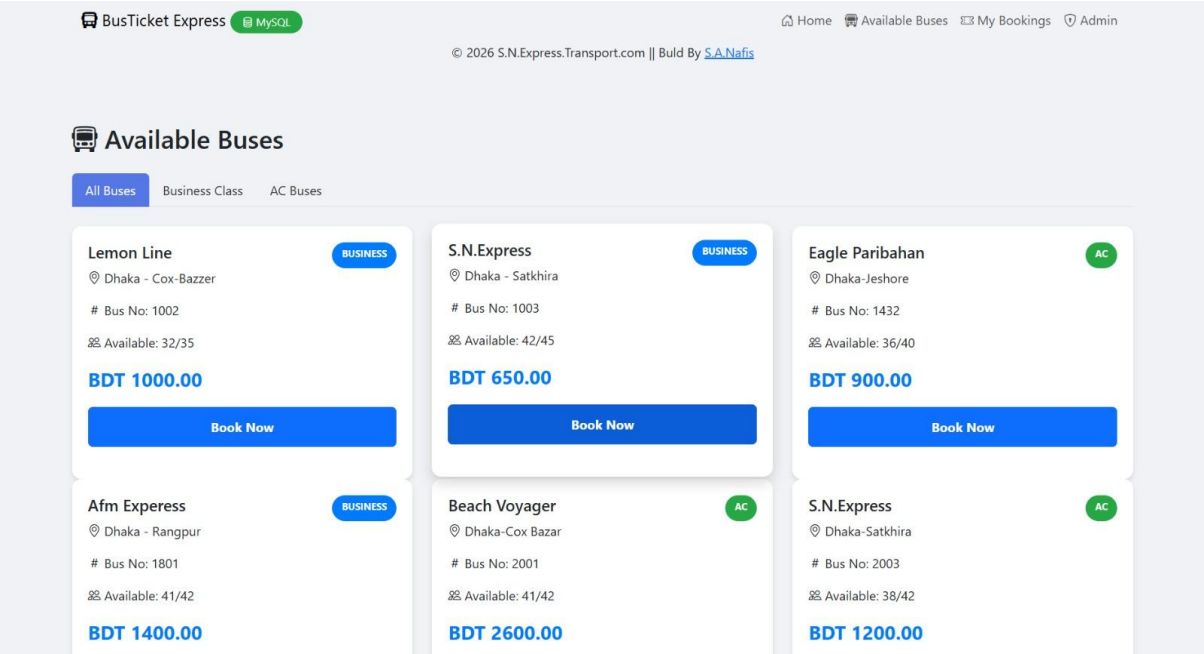
```
        cursor.execute("SELECT bus_name, total_seats, available_seats FROM buses WHERE
bus_no = %s", (bus_no,))
        bus = cursor.fetchone()

        if not bus:
            return jsonify({"error": "Bus not found"}), 404

        cursor.execute("""
            UPDATE buses
            SET available_seats = total_seats
            WHERE bus_no = %s
        """, (bus_no,))

        conn.commit()

        return jsonify({
            "message": f"□ Bus #{bus_no} ({bus['bus_name']}) reset to {bus['total_seats']} seats!",
            "bus_no": bus_no,
            "bus_name": bus['bus_name'],
            "total_seats": bus['total_seats']
        }), 200

    except Exception as e:
        conn.rollback()
        return jsonify({"error": str(e)}), 500
    finally:
```

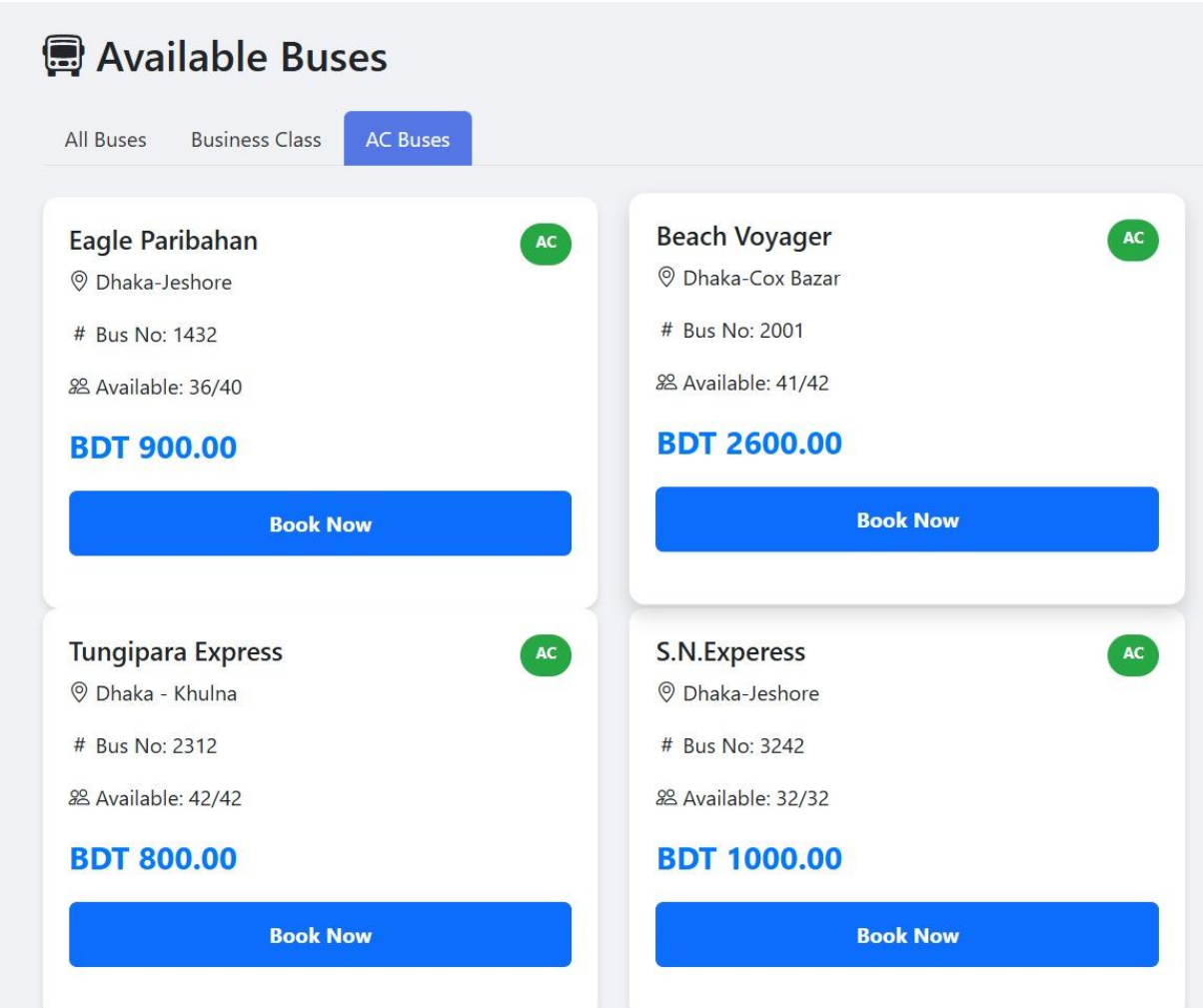
```
    cursor.close()
    conn.close()

if __name__ == '__main__':
    print("\n" + "="*50)
    print("🚌 Bus Booking System - MySQL Connected")
    print("="*50)
    print(f"📍 Frontend Directory: {FRONTEND_DIR}")
    print("📍 http://localhost:5000")
    print("🗄️ Database: bus_booking_system")
    print("="*50 + "\n")
    app.run(debug=True, host='0.0.0.0', port=5000)
```

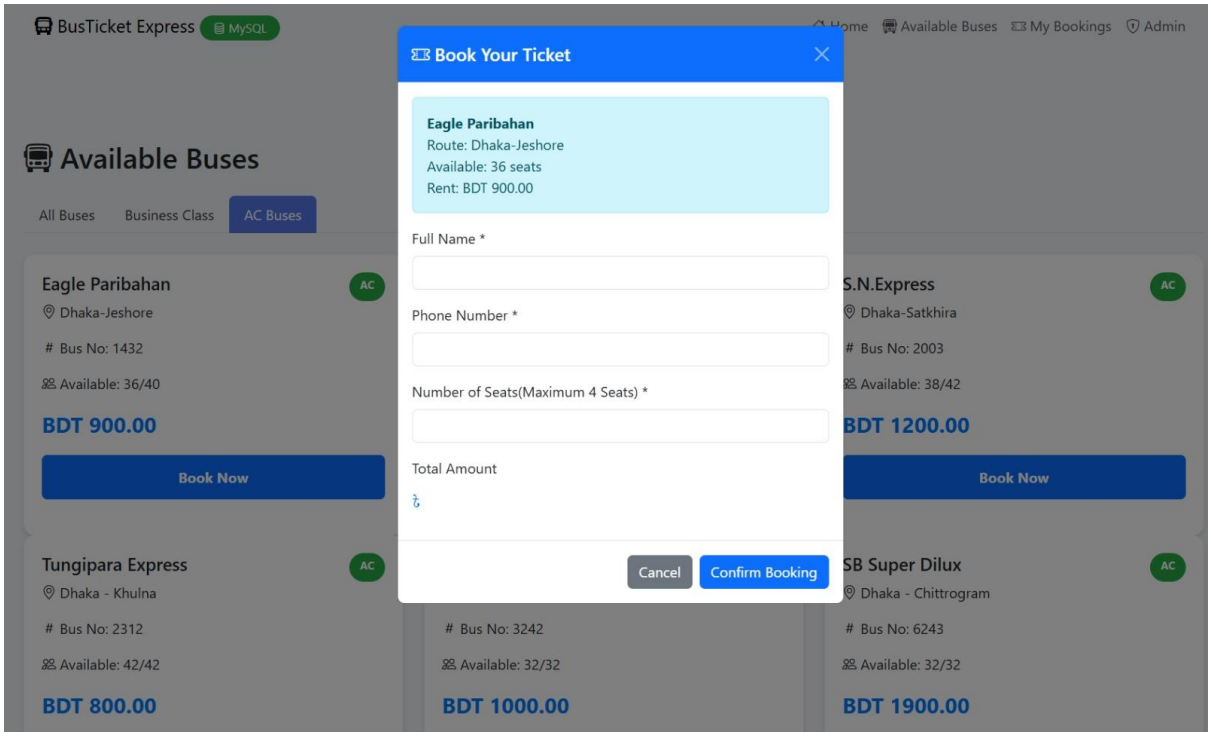
OUTPUT

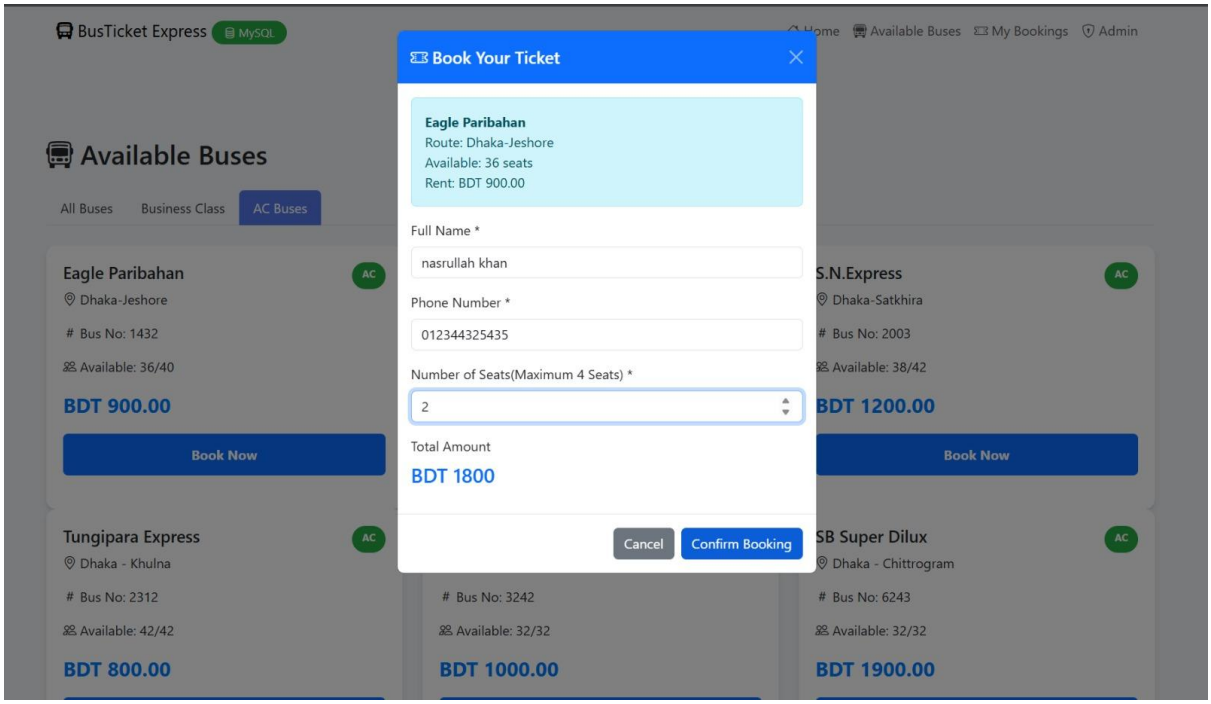


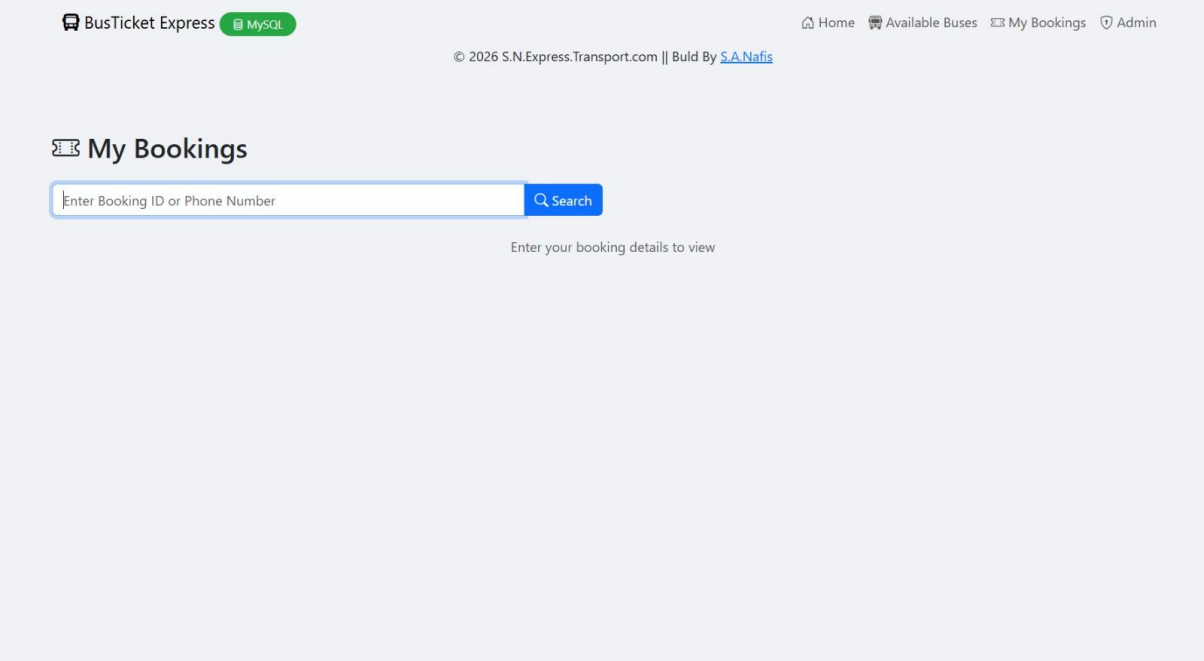

BusTicket Express
MySQL
Home
Available Buses
My Bookings
Admin
© 2026 S.N.Express.Transport.com || Buld By S.A.Nafis
Available Buses
All Buses
Business Class
AC Buses
Lemon Line
BUSINESS
Dhaka - Cox-Bazzer
Bus No: 1002
Available: 32/35
BDT 1000.00
Book Now
S.N.Express
BUSINESS
Dhaka - Satkhira
Bus No: 1003
Available: 42/45
BDT 650.00
Book Now
Eagle Paribahan
AC
Dhaka-Jeshore
Bus No: 1432
Available: 36/40
BDT 900.00
Book Now
Afm Experess
BUSINESS
Dhaka - Rangpur
Bus No: 1801
Available: 41/42
BDT 1400.00
Beach Voyager
AC
Dhaka-Cox Bazar
Bus No: 2001
Available: 41/42
BDT 2600.00
S.N.Express
AC
Dhaka-Satkhira
Bus No: 2003
Available: 38/42
BDT 1200.00

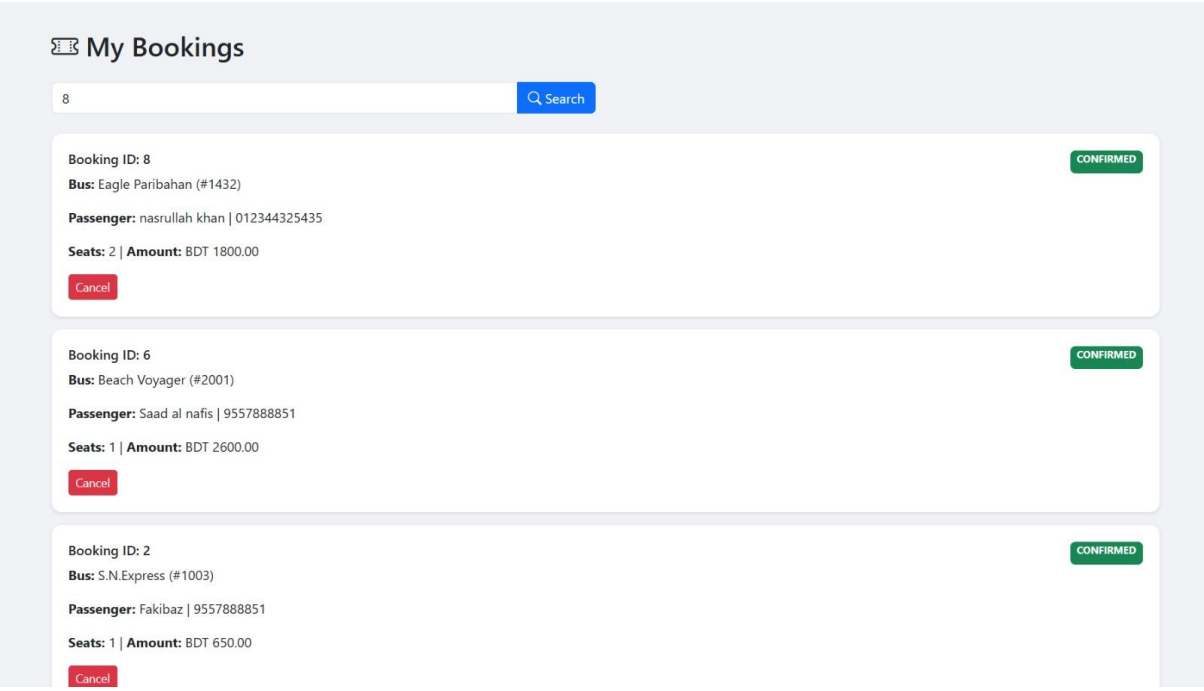

BusTicket Express
MySQL
Home
Available Buses
My Bookings
Admin
© 2026 S.N.Express.Transport.com || Buld By S.A.Nafis
Available Buses
All Buses
Business Class
AC Buses
Lemon Line
BUSINESS
Dhaka - Cox-Bazzer
Bus No: 1002
Available: 32/35
BDT 1000.00
Book Now
S.N.Express
BUSINESS
Dhaka - Satkhira
Bus No: 1003
Available: 42/45
BDT 650.00
Book Now
Eagle Paribahan
AC
Dhaka-Jeshore
Bus No: 1432
Available: 36/40
BDT 900.00
Book Now
Afm Experess
BUSINESS
Dhaka - Rangpur
Bus No: 1801
Available: 41/42
BDT 1400.00
Beach Voyager
AC
Dhaka-Cox Bazar
Bus No: 2001
Available: 41/42
BDT 2600.00
S.N.Express
AC
Dhaka-Satkhira
Bus No: 2003
Available: 38/42
BDT 1200.00


Available Buses
All Buses
Business Class
AC Buses
Eagle Paribahan
AC
Dhaka-Jeshore
Bus No: 1432
Available: 36/40
BDT 900.00
Book Now
Beach Voyager
AC
Dhaka-Cox Bazar
Bus No: 2001
Available: 41/42
BDT 2600.00
Book Now
Tungipara Express
AC
Dhaka - Khulna
Bus No: 2312
Available: 42/42
BDT 800.00
Book Now
S.N.Experess
AC
Dhaka-Jeshore
Bus No: 3242
Available: 32/32
BDT 1000.00
Book Now


BusTicket Express
MySQL
Home
Available Buses
My Bookings
Admin
Available Buses
All Buses
Business Class
AC Buses
Eagle Paribahan
AC
Dhaka-Jeshore
Bus No: 1432
Available: 36/40
BDT 900.00
Book Now
Tungipara Express
AC
Dhaka - Khulna
Bus No: 2312
Available: 42/42
BDT 800.00
Bus No: 3242
Available: 32/32
BDT 1000.00
S.N.Express
AC
Dhaka-Satkhira
Bus No: 2003
Available: 38/42
BDT 1200.00
Book Now
SB Super Dilux
AC
Dhaka - Chittrogram
Bus No: 6243
Available: 32/32
BDT 1900.00
Book Your Ticket
Eagle Paribahan
Route: Dhaka-Jeshore
Available: 36 seats
Rent: BDT 900.00
Full Name *
Phone Number *
Number of Seats(Maximum 4 Seats) *
Total Amount
৳
Cancel
Confirm Booking


BusTicket Express
MySQL
Home
Available Buses
My Bookings
Admin
Available Buses
All Buses
Business Class
AC Buses
Eagle Paribahan
AC
Dhaka-Jeshore
Bus No: 1432
Available: 36/40
BDT 900.00
Book Now
S.N.Express
AC
Dhaka-Satkhira
Bus No: 2003
Available: 38/42
BDT 1200.00
Book Now
Tungipara Express
AC
Dhaka - Khulna
Bus No: 2312
Available: 42/42
BDT 800.00
Bus No: 3242
Available: 32/32
BDT 1000.00
SB Super Dilux
AC
Dhaka - Chittrogram
Bus No: 6243
Available: 32/32
BDT 1900.00
Book Your Ticket
Eagle Paribahan
Route: Dhaka-Jeshore
Available: 36 seats
Rent: BDT 900.00
Full Name *
nasrullah khan
Phone Number *
012344325435
Number of Seats(Maximum 4 Seats) *
2
Total Amount
BDT 1800
Cancel
Confirm Booking


BusTicket Express
MySQL
Home
Available Buses
My Bookings
Admin
© 2026 S.N.Express.Transport.com || Buld By S.A.Nafis
My Bookings
Enter Booking ID or Phone Number
Search
Enter your booking details to view



CONCLUSION

The Bus Ticket Express project successfully delivers a fully functional web-based bus ticketing system. Passengers can browse available buses, filter by bus type, book tickets online, and cancel bookings when needed — all from a single platform.

The system includes an admin panel that allows bus operators to add or remove buses as required. All data is stored and managed using a MySQL database, ensuring real-time seat availability and accurate booking records.

This project demonstrates how a simple, user-friendly website can replace traditional manual ticketing systems. It saves time for passengers, reduces the need to visit physical counters, and provides a convenient digital solution for bus ticket booking.